Arista Access Node MAR-ACC-45 - QoS Congestion Event Root Cause Analysis and Resolution SOP

Node: MAR-ACC-45 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6301

Description

QoS congestion events on Arista access nodes, such as MAR-ACC-45 located at the Marathahalli Core site, are often caused by microbursts or interface tail drops resulting in QoS queue starvation. This is due to the inherent design of the Arista's egress queue management, which relies on the priority-based queuing (PQ) mechanism. In a high-traffic environment like the Marathahalli Core site, where multiple concurrent flows compete for bandwidth, the PQ algorithm may struggle to efficiently manage bandwidth allocation leading to congestion events. Moreover, the Arista's egress queue management is also influenced by the tail-drop policy, which discards packets when the queue is full, further exacerbating congestion. Therefore, it is crucial to investigate and resolve QoS congestion events on access nodes like MAR-ACC-45 to prevent network degradation and ensure Quality of Service.

Pre-requisite Checks

- Verify that the redundant link is operational and functioning correctly.

- Confirm that the site's environmental conditions are within the standard operating temperature and humidity range.

- Check the power supply to the node and ensure it is functioning correctly.

- Verify that the node's firmware is up-to-date and the latest version is installed.

Potential Root Causes

Cause	Probability
Inefficient PQ algorithm leading to QoS queue starvation	High
Tail-drop policy causing packet discard and congestion	Medium
Misconfigured QoS policies or incorrect priority settings	Low

Diagnostic Commands & Expected Outputs

```
Command: show queueing interface, Expected Output: Queueing information for interface Ethernet1/1: ..., Description: This command displays the current queueing configuration and utilization for the interface, helping to identify potential congestion points.

Command: show qos policy, Expected Output: Policy name: qos_policy1, Description: This command displays the current QoS policy configuration, allowing us to verify if the policy is correctly configured and if the priority settings are correct.
```

Resolution Steps

Step 1: Step: Optimize PQ algorithm configuration, Command: configure terminal queuing pq interface Ethernet1/1 queue 1 bandwidth 50 end, Description: This command adjusts the PQ algorithm configuration to allocate more bandwidth to the queue, helping to prevent QoS queue starvation.

Step 2: Step: Adjust tail-drop policy, Command: configure terminal tail-drop policy interface Ethernet1/1 drop-threshold 100 end, Description: This command adjusts the tail-drop policy to increase the drop threshold, reducing the likelihood of packet discard and congestion.

Post-Check Verification

```
Command: show queueing interface, Expected Output: Queueing
information for interface Ethernet1/1: ... (optimized configuration),
Description: This command verifies that the optimized PQ algorithm
configuration is in effect and the queueing configuration has been
adjusted correctly.

Command: show qos policy, Expected Output: Policy name: qos_policy1
(correctly configured), Description: This command verifies that the
QoS policy configuration has been corrected and the priority settings
are correct.
```

Roll-back Procedures

- Step: Restore original PQ algorithm configuration, Command: configure terminal undo queuing pq interface Ethernet1/1 queue 1 bandwidth 50 end, Description: This command restores the original PQ algorithm configuration, reverting any changes made during the resolution process.

- Step: Restore original tail-drop policy, Command: configure terminal tail-drop policy interface Ethernet1/1 drop-threshold 50 end, Description: This command restores the original tail-drop policy, reverting any changes made during the resolution process.